Jian Wu, Ph.D.

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Employment History

2024 – · · · · **Asistant Researcher,** State Key Laboratory of Virtual Reality Technology and Systems, Beihang University, Beijing, China.

Education

2018 – 2024 **Ph.D. Computer Science, Beihang University** in Computer Application Technology. Thesis title: *Research on Interactive Presentation Methods for Multi-scale Virtual Scenes.*

2015 – 2018 **M.Sc. Mechanical Engineering, Beihang University** in Mechanical Design. Thesis title: *Research on Lightweight Topology Optimization Design Method of Typical Structure Based on Additive Manufacturing.*

2011 – 2015 **B.E. Mechanical Engineering and Automation, Beihang University** in Mechanical Engineering and Automation.

Research Publications

Journal Articles

1. S. Luan, J. Wu, D. Zhou, W. Ke, and L. Wang, "User perception based label layout for efficient target localization in virtual environment," *International Journal of Human-Computer Studies*, p. 103 801, 2026.
2. L. Wang et al., "Artificial intelligence for virtual reality: A review," *Science China Information Sciences*, vol. 69, no. 1, p. 111 101, 2026.
3. J. Wu et al., "Align-box: Enable fast bare-hand multi-object alignment in vr," *International Journal of Human–Computer Interaction*, pp. 1–23, 2026.
4. J. Wu et al., "Automatic formation generation based on scene awareness for guided group navigation in vr," *IEEE Transactions on Visualization and Computer Graphics*, 2026.
5. R. Fan, J. Wu, X. Shi, L. Zhao, Q. Ma, and L. Wang, "Fov-gs: Foveated 3d gaussian splatting for dynamic scenes," *IEEE Transactions on Visualization and Computer Graphics*, 2025.
6. S. Huang, J. Wu, R. Fan, S. K. Im, and L. Wang, "Handbrush for efficient object grouping in virtual environment with bare-hand," *International Journal of Human–Computer Interaction*, vol. 41, no. 15, pp. 9797–9821, 2025.
7. Q. Ma, R. Fan, L. Zhao, J. Wu, S.-K. Im, and L. Wang, "Sgsg: Stroke-guided scene graph generation," *IEEE Transactions on Visualization and Computer Graphics*, 2025.
8. X. Shi et al., "Audio-visual aware foveated rendering," *IEEE Transactions on Visualization and Computer Graphics*, 2025.
9. X. Shi et al., "Moa: Efficient scene-aware multi-object arrangement in vr," *IEEE Transactions on Visualization and Computer Graphics*, 2025.
10. K. Wang, J. Wu, R. Fan, H. Zhang, S. K. Im, and L. Wang, "Hfm-gs: Half-face mapping 3dgs avatar based real-time hmd removal," *IEEE Transactions on Visualization and Computer Graphics*, 2025.

11 L. Wang, X. Li, J. Wu, D. Zhou, I. Sio Kei, and V. Popescu, "Avicol: Adaptive visual instruction for remote collaboration using mixed reality," *International Journal of Human–Computer Interaction*, vol. 41, no. 2, pp. 1260–1279, 2025.

12 P. Wang, M. Li, J. Wu, L. Wang, and Y.-J. Liu, "Cybersickness exploration for different vr tasks under variable rendering conditions," *International Journal of Human–Computer Interaction*, pp. 1–24, 2025.

13 J. Wu, W. Zhang, H. Chen, W. Lin, X. Shi, and L. Wang, "Pwp: Permutating with probability for efficient group selection in vr," *IEEE Transactions on Visualization and Computer Graphics*, 2025.

14 J. Zhou, J. Wu, S. K. Im, R. Fan, and L. Wang, "Manipulable cone based bare hand object selection in high occlusion virtual environment," *International Journal of Human-Computer Studies*, vol. 196, p. 103 432, 2025.

15 X. Liu, L. Wang, Y. Liu, and J. Wu, "Automatic portals layout for vr navigation," *Virtual Reality*, vol. 28, no. 1, p. 9, 2024.

16 Z. Liu, J. Wu, L. Wang, X. Li, and S. K. Im, "Proxy importance based haptic retargeting with multiple props in vr," *IEEE Transactions on Visualization and Computer Graphics*, 2024.

17 X. Shi, L. Wang, X. Liu, J. Wu, and Z. Shao, "Scene-aware foveated neural radiance fields," *IEEE Transactions on Visualization and Computer Graphics*, 2024.

18 A. Sun, J. Wu, R. Fan, S. K. Im, and L. Wang, "Efficient and comfortable haptic retargeting with reset point optimization," *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 9, pp. 6588–6602, 2024.

19 Z. Wang, J. Wu, R. Fan, W. Ke, and L. Wang, "Vprf: Visual perceptual radiance fields for foveated image synthesis," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 11, pp. 7183–7192, 2024.

20 J. Wu and L. Wang, "Panoramic ray tracing for interactive mixed reality rendering based on 360° rgbd video," *IEEE Computer Graphics and Applications*, vol. 44, no. 1, pp. 62–75, 2023.

21 Z. Zhao, J. Wu, and L. Wang, "Ar assistance for efficient dynamic target search," *Computational Visual Media*, vol. 9, no. 1, pp. 177–194, 2023.

22 X. Shi, L. Wang, J. Wu, R. Fan, and A. Hao, "Foveated stochastic lightcuts," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 11, pp. 3684–3693, 2022.

23 J. Wu, L. Wang, H. Zhang, and V. Popescu, "Quantifiable fine-grain occlusion removal assistance for efficient vr exploration," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 9, pp. 3154–3167, 2021.

24 L. Wang, J. Wu, X. Yang, and V. Popescu, "Vr exploration assistance through automatic occlusion removal," *IEEE transactions on visualization and computer graphics*, vol. 25, no. 5, pp. 2083–2092, 2019.

Conference Proceedings

1 J. Leng, Z. Wang, J. Wu, and L. Wang, "Liptext: Lip tracking based text entry in vr," in *International Conference on Extended Reality*, Springer, 2024, pp. 162–178.

2 X. Liu, K. Jiang, J. Wu, L. Wang, and G. Geng, "Where should a virtual guide stand in a vr museum?" In *International Conference on Extended Reality*, Springer, 2024, pp. 312–328.

3 Q. Ma, J. Wu, R. Fan, G. Sun, and X. Shi, "Vip-fluid: Visual perception driven method for vr fluid rendering," in *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, IEEE, 2024, pp. 359–367.

4 J. Wu, Z. Wang, L. Wang, Y. Duan, and J. Li, "Fanpad: A fan layout touchpad keyboard for text entry in vr," in *2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, IEEE, 2024, pp. 222–232.

5 X. Shi, L. Wang, J. Wu, W. Ke, and C.-T. Lam, "Locomotion-aware foveated rendering," in *2023 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*, IEEE, 2023, pp. 471–481.

6 J. Wu, L. Wang, and W. Ke, "Interactive panoramic ray tracing for mixed 360° rgbd videos," in *2023 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, IEEE, 2023, pp. 777–778.

7 L. Wang, Y. Liu, X. Liu, and J. Wu, "Automatic virtual portals placement for efficient vr navigation," in *2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, IEEE, 2022, pp. 628–629.

8 L. Wang et al., "Occlusion management in vr: A comparative study," in *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, IEEE, 2019, pp. 708–716.

Skills

Languages	Strong reading, writing and speaking competencies for English, Japanese.
Coding	C++, Python, SQL, XML/XSL, LaTeX, …
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Miscellaneous Experience

Awards and Achievements

2025 **Best Paper Honorable Mention**, IEEE VR 2025.

2024 **Best Student Journal Paper**, IEEE ISMAR 2024.